Determine the Area of Parallelograms and Rhombuses

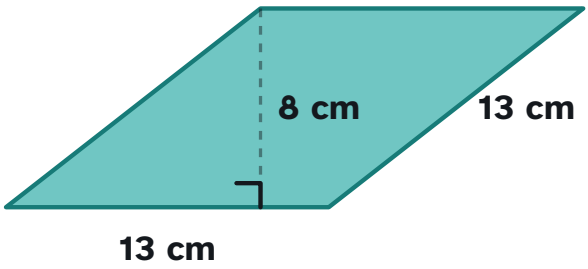
Find the area of a rhombus. It is a parallelogram, so use base × straight-up height.

Name _____ **Date** _____

Tonight's spark

A hardware store sells diamond-shaped floor tiles. The sign says every side is 13 cm. A ruler straight up from the bottom edge to the top edge reads 8 cm. How much floor does one tile cover?

Guess first: is it more or less than 150 square centimeters?



Our class model: a rhombus is a parallelogram

Try it

Work it out on paper. Write one answer on each line.

1 **Which statement about a rhombus is true?**

A. It has no parallel sides **B.** It is a parallelogram with four equal sides
C. It has only two equal sides **D.** Its height is always the same as its side

My answer _____

2 **A rhombus has a base of 7 cm and a height of 5 cm. What is its area?**

A. 12 cm^2 **B.** 28 cm^2 **C.** 17.5 cm^2 **D.** 35 cm^2

My answer _____

3 **A rhombus has sides of 10 in. and a straight-up height of 8 in. What is its area?**

A. 80 in² **B.** 100 in² **C.** 18 in² **D.** 40 in²

My answer _____

4 **A rhombus has four sides of 15 cm. A segment from the top side straight down to the base, at a right angle, measures 9 cm. Which number is the height?**

A. 15, the side printed on the picture **B.** 24, the two numbers added

C. 9, the straight-up distance **D.** 6, the difference of the two numbers

My answer _____

5 **A rhombus-shaped sign has a base of 12 in. and a height of 9 in. What is its area?**

A. 21 in² **B.** 108 in² **C.** 54 in² **D.** 144 in²

My answer _____

Talk about it

A rhombus has four equal sides. Which measurement is its height, and why is the side not the height?

Home mission

A different situation: lay four pencils of the same length into a square. Gently push one corner so it leans into a diamond. The sides stay the same. Which measure got smaller: the side, or the straight-up height?

If we get stuck, ask

- Is that measurement straight up from the base, or slanted?
- A rhombus is which bigger family of shapes?
- What unit does an area answer use?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/5-1-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it